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Q1. Newton's second law: rate of change of momentum proportional to force, in direction of force. $F=ma$.

Q1. (contd) Also this is sometimes written as $F = dp/dt$ where p is momentum.

Q3. Speed is scalar, velocity is vector. Speed = 40km/hr, Velocity = 40km/hr North. ~~(this was actually Q2 sorry)~~

Q4. SI unit of current is Ampere. One ampere = one coulomb per second.

Q6 (numerical). $u=0$, $a=2$, $t=5s$. $v=u+at=10m/s$. $s=ut+\frac{1}{2}at^2=25m$.

Q8. Ohm's law $V=IR$. $V=2 \times 10=20$ Volts.

Q.11 part a $v=u+at$, avg velocity $= (u+v)/2$, $s=avg \times t$, so $s=ut+\frac{1}{2}at^2$

Q9 Three states of matter: solid, liquid, gas – particle spacing differs in each.

Q7. Mitosis – 2 identical cells for growth. Meiosis – 4 cells, half chromosomes, for reproduction.

Q13. (no such question on paper) Two ways to conserve water: rainwater harvesting and fixing leaks.